MATE70020
Modelling Materials with Density Functional Theory

2026 Fall

Lecture 3

PSet 1 due today at 1500
PSet 2 is released and due Nov 10 (in two weeks)

Modelling monolayer boron nitride

Recap

Many-body Schrödinger's equation

$$\hat{H}\,\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N) = E\,\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N)$$

$$\hat{H} = \sum_{i=1}^{N} -\frac{\hbar^2}{2m}\nabla_i^2 + \sum_{i=1}^{N} V(\vec{r}_i) + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi|\vec{r}_i - \vec{r}_j|} + \frac{1}{2}\sum_{I\neq J}\frac{e^2}{4\pi|\vec{R}_I - \vec{R}_J|}$$

Variational principle: turn solving the equation into minimizing energy

$$F[\Psi] \geq E_0$$

Hohenberg-Kohn theorem: use the electron density $n(\vec{r})$ as the variable instead of the many-body wavefunction

$$F[n] \geq E_0$$

but don't know form of $F[n]$

Kohn-Sham equation: use non-interacting electrons instead of the real interacting system

$$F[n] = \sum_{i=1}^{N} - \int dV\, \psi_i^*(\vec{r}) \frac{\hbar^2 \nabla^2}{2m} \psi_i(\vec{r}) + \int dV\, n(\vec{r}) V(\vec{r})$$

$$+ \frac{e^2}{8\pi\varepsilon_0} \int dV \int dV' \frac{n(\vec{r})\, n(\vec{r}')}{|\vec{r} - \vec{r}'|} + E_{xc}[n]$$

exchange correlation

Apply variational principle

$$0 = \frac{\delta}{\delta \psi_j^*} \left[F[n] - \sum_j \lambda_j \left(\int dV' |\psi_j(\vec{r}')|^2 - 1 \right) \right]$$

$$\frac{\delta}{\delta \psi_i^*} \sum_j \int dV' \psi_j^*(\vec{r}') \left[-\frac{\hbar^2}{2m} \nabla^2 \psi_j(\vec{r}') \right] = -\frac{\hbar^2}{2m} \nabla^2 \psi_i(\vec{r})$$

$\delta_{ij}\, \delta(\vec{r} - \vec{r}')$

$= 0$

$$\frac{\delta}{\delta \psi_i^*} \int dV' \underbrace{\sum_j \psi_j^*(\vec{r}')\,\psi_j(\vec{r}')}_{n(\vec{r}')}\, V(\vec{r}') = V(\vec{r})\,\psi_i(\vec{r})$$

Hartree term

$$= \underbrace{\frac{e^2}{8\pi\varepsilon_0} \int dV' \frac{n(\vec{r}')}{|\vec{r}-\vec{r}'|}}_{V_H(\vec{r})}\, \psi_i(\vec{r})$$

$$= V_H(\vec{r})\,\psi_i(\vec{r})$$

XC:

$$\frac{\delta}{\delta \psi_i^*} E_{xc}[n] = \underbrace{\frac{\delta E_{xc}}{\delta n(\vec{r})}}_{v_{xc}(\vec{r})} \psi_i(\vec{r})$$

KS equation

$$-\frac{\hbar^2}{2m}\nabla^2 \psi_i(\vec{r}) + V(\vec{r})\psi_i(\vec{r}) + v_H(\vec{r})\psi_i(\vec{r}) + v_{xc}(\vec{r})\psi_i(\vec{r}) = \varepsilon_i \psi_i(\vec{r})$$

$$-\frac{\hbar^2}{2m}\nabla^2 \psi_i(\vec{r}) + V_{KS}(\vec{r})\psi_i(\vec{r}) = \varepsilon_i \psi_i(\vec{r})$$

Note: $\psi_i(\vec{r})$ and ε_i do not have physical significance but can calculate $n(\vec{r})$ and E_c

Lecture 3: Practical aspects of DFT

The exchange-correlation functional
Solving the Kohn-Sham equation on a computer
Pseudopotentials

The exchange-correlation functional

E_{xc} exact form is not known

simplest approximation: local density approximation (LDA)

model system: uniform electron gas in free space

$V(\vec{r}) = \text{const} \Rightarrow n(\vec{r}) = \text{const} = n$

For this system: possible to solve many-el S.E.

Quantum Monte Carlo

$\Rightarrow E_0$ as a function of n

E_0: kinetic energy + Hartree energy + XC

subtract the kinetic and Hartree energy

$\rightarrow \epsilon_{xc}^{FS}(n)$

Real Materials

LDA assumes E_{xc} of electrons in volume dV at $\vec{r}$ is the same as if those electrons were in a uniform electron gas with constant density $n = n(\vec{r})$

$$E_{xc}[n(\vec{r})] = \int dV \epsilon_{xc}^{FS}(n(\vec{r}))$$

Beyond LDA

generalized gradient approximation (GGA)

$$E_{xc}[n, \nabla n]$$

meta-GGA

$$E_{xc}\left[n, \nabla n, \tau = \sum_i \frac{1}{2} |\nabla \psi_i(\vec{r})|^2\right]$$

LDA works well for:

- material with strong bonds (covalent, ionic, metallic)
- predictions of bond length, elastic constants, vibrations

Not accurate for:

- band gaps
- weakly bonded materials

Hybrid functionals

- separate E_{xc} in E_x and E_c

replace a fraction of E_x with Hartree-Fock E_x

$$E_x^{HF} = -\sum_{ij} \int dV \int dV' \frac{\psi_i^*(\vec{r})\,\psi_j(\vec{r})\,\psi_j^*(\vec{r})\,\psi_i(\vec{r})}{|\vec{r}-\vec{r}'|}$$

$$E_{xc} = E_c^{DFT} + (1-\alpha)\,E_x^{DFT} + \alpha\,E_x^{HF}$$

α: mixing parameter

PBE0 $\alpha = 0.25$ default

better for band gaps

much more expensive

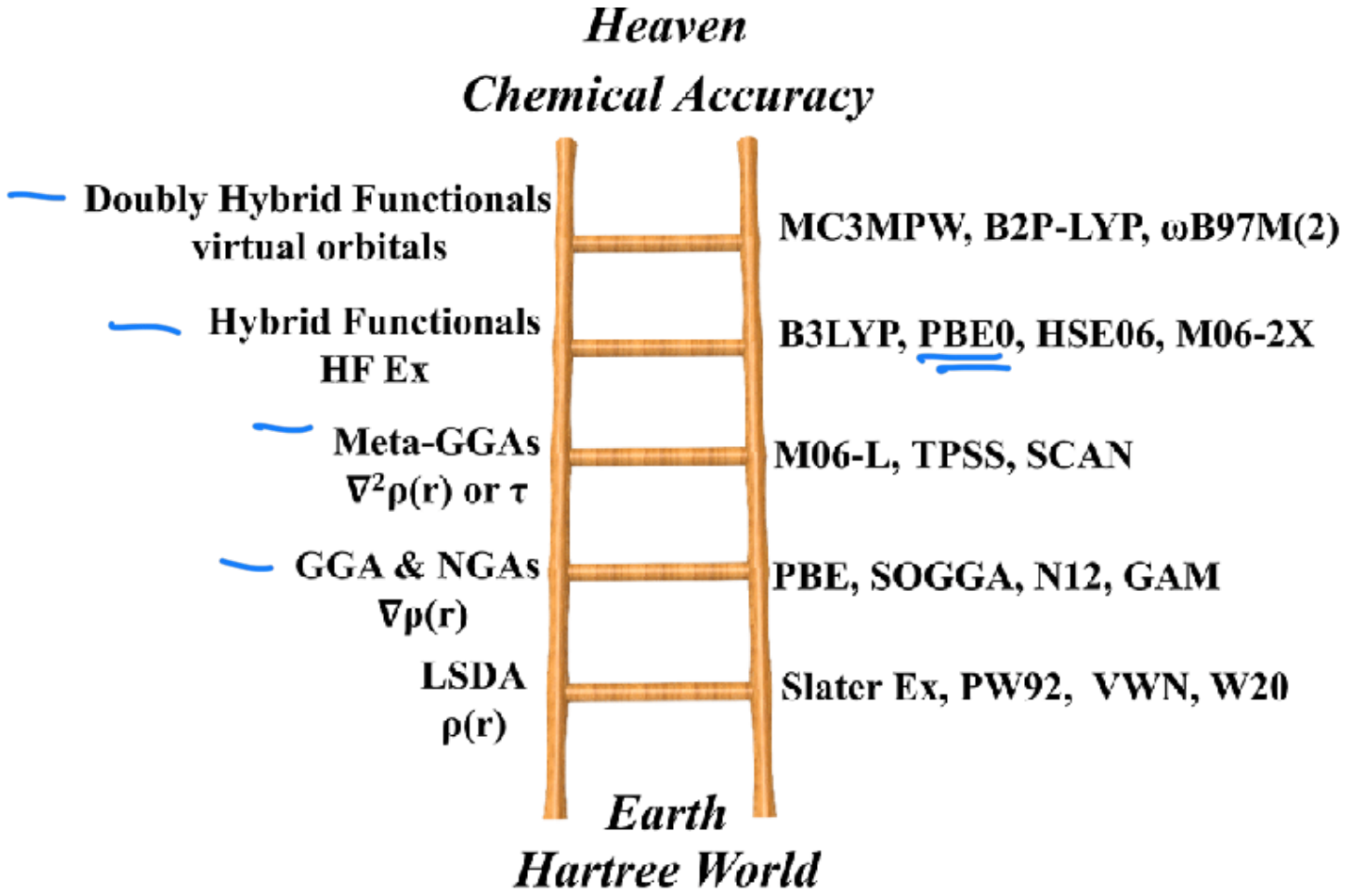
van der Waals functionals

- important for layered materials, molecular crystals, biomaterial

- Grimme D3 correction

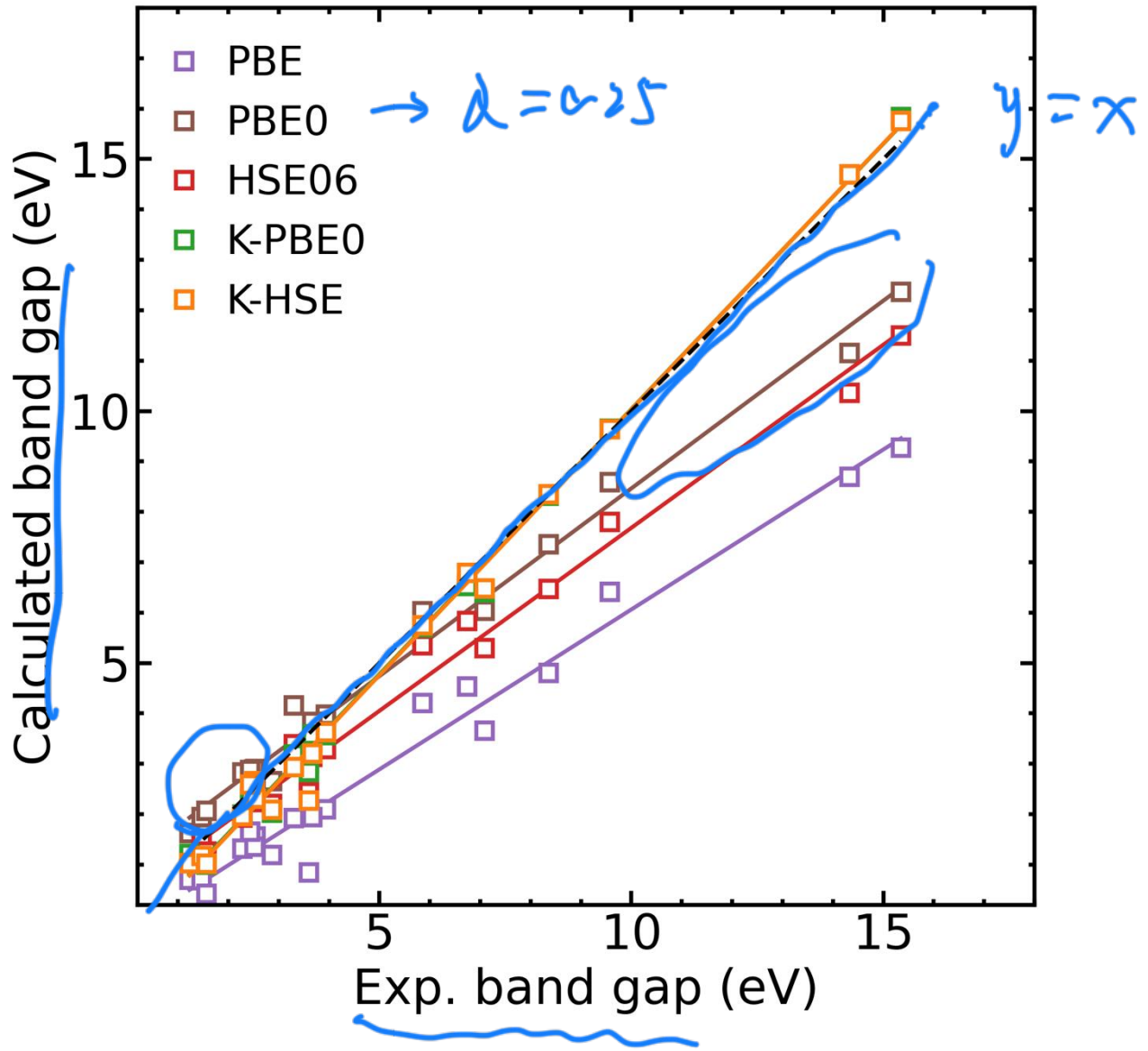
- vdW-DF

Jacob's ladder



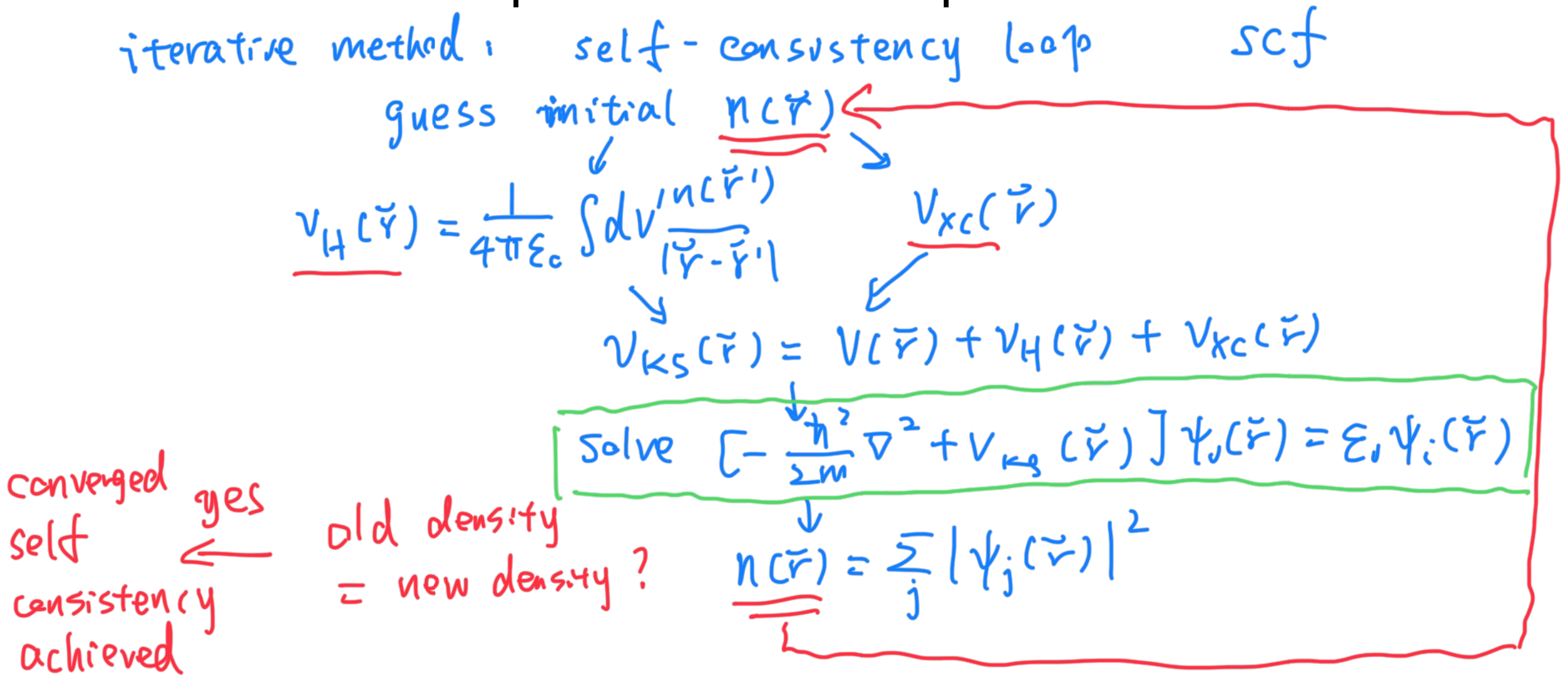
Energy Environ. Mater. 2021, 0,1–29

	PBE0	Expt.+ZPR
sp materials		
AlN	6.28	6.71
AlP	2.92	2.60
AlAs	2.71	2.31
Ar	11.14	14.33
BN	6.57	6.74[a]
C (diamond)	6.05	5.85[a]
CaO	6.03	7.43
LiCl	8.58	9.57
LiF	12.31	15.43
MgO	7.25	8.47
Si	1.76	1.22
SiC	2.91	2.59
NaCl	7.14	9.14
MAE_{sp}	1.10	
$MARE_{sp}$	15.9%	
3*d* materials		
Ge	1.35	0.78
GaN	3.84	3.73
GaP	2.93	2.41
GaAs	1.95	1.57
InP	2.10	1.47
ZnO	3.19	3.61
ZnS	3.99	3.95
ZnSe	2.95	2.88
TiO_2	4.18	3.65
Cu_2O	2.66	2.21
CdS	2.89	2.66
CdSe	2.34	1.88
γ-$CsSnI_3$	1.75	1.53
MAE_d	0.36	
$MARE_d$	20.1%	
MAE	0.73	
MARE	18.0%	



J. Yang et al., npj Computational Materials (2023) 9:108

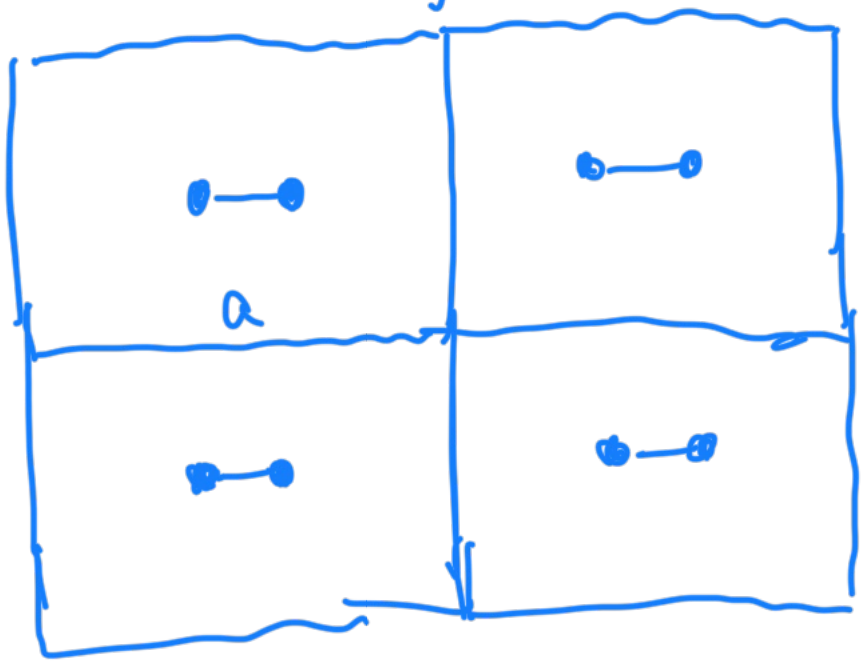
How to solve KS equation on a computer?



Plane-wave based method

Consider N_2 molecule

periodic boundary condition



$\Rightarrow$ not satisfied for solids

$$\psi(\vec{r} + a\hat{x}) = \psi(\vec{r})$$

Fourier expansion (plane wave)

$$\psi(\vec{r}) = \sum_{\vec{G}} C_{\vec{G}} e^{i\vec{G}\cdot\vec{r}}$$

What values can $\vec{G}$ take?

$$\sum_{\vec{G}} C_{\vec{G}} e^{i\vec{G}\cdot(\vec{r}+a\hat{x})} = \sum_{\vec{G}} C_{\vec{G}} e^{i\vec{G}\cdot\vec{r}}$$

$$\Rightarrow e^{i\vec{G}\cdot a\hat{x}} = 1$$

$$\vec{G}\cdot a\hat{x} = 2\pi n_x$$

$$G_x = \frac{2\pi}{a_x} n_x$$

$$\textcircled{1}\ -\frac{\hbar^2}{2m}\nabla^2\psi_i + \textcircled{2}\ V_{KS}(\vec{r})\,\psi_i = \textcircled{3}\ \epsilon_i\,\psi_i$$

$$V_{KS} = \sum_{\vec{G}} V_{\vec{G}}\, e^{i\vec{G}\cdot\vec{r}} \qquad \psi = \sum_{\vec{G}} C_{\vec{G}}\, e^{i\vec{G}\cdot\vec{r}}$$

Plug into KS equation

Term ① $\sum_{\vec{G}} \frac{\hbar^2}{2m}\vec{G}^2 C_{\vec{G}}\, e^{i\vec{G}\cdot\vec{r}}$

Term ② $\sum_{\vec{G}}\sum_{\vec{G}'} V_{\vec{G}} C_{\vec{G}'}\, e^{i(\vec{G}+\vec{G}')\cdot\vec{r}}$

$\vec{G}'' = \vec{G} + \vec{G}' \qquad \vec{G} = \vec{G}'' - \vec{G}'$

$$= \sum_{\vec{G}''}\sum_{\vec{G}''-\vec{G}'} C_{\vec{G}'} V_{\vec{G}''-\vec{G}'}\, e^{i\vec{G}''\cdot\vec{r}}$$

rewrite $\vec{G}''$ to $\vec{G}$

$$= \sum_{\vec{G}}\sum_{\vec{G}-\vec{G}'} C_{\vec{G}'} V_{\vec{G}-\vec{G}'}\, e^{i\vec{G}\cdot\vec{r}}$$

Term ③ $\epsilon \sum_{\vec{G}} C_{\vec{G}}\, e^{i\vec{G}\cdot\vec{r}}$

$$\left(\frac{\hbar^2}{2m}G^2 - \varepsilon\right) c_{\vec{G}} + \sum_{\vec{G}'} V_{\vec{G}-\vec{G}'} c_{\vec{G}'} = 0$$

$H\vec{c} = \varepsilon\vec{c}$ $\vec{c}$ is a vector of $c_{\vec{G}}$

matrix ergen value problem

Introduce a cut-off

only keep $\frac{\hbar G^2}{2m} < E_{cut}$

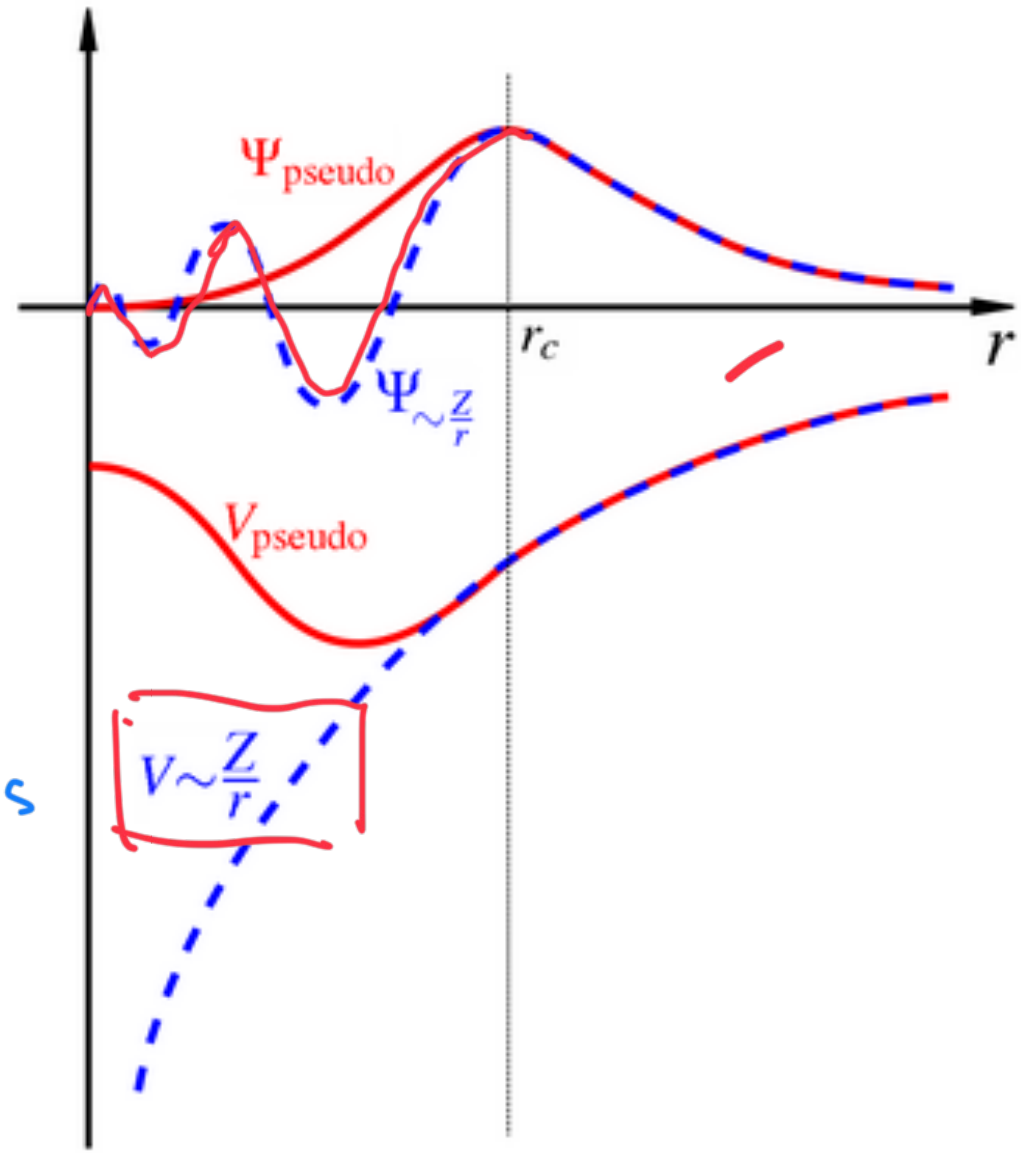
Convergence parameter

Pseudopotential

Idea: combine nucleus and core electrons into effective nucleus whose effect on the valence electrons is described by pseudopotential

Advantage:

- need to deal with fewer els
- can use lower cut-off



How to construct ?

- for isolated atom: calculate KS wavefunctions and KS energies with all electrons

$\psi_n^{AE}(\vec{r})$, ε_n^{AE}

- solve KS equation for isolated atom for valence electrons of effective nucleus

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V_H(\vec{r}) + V_{xc}(\vec{r}) + V_{PP}(\vec{r})\right]\psi_n^{PP}(\vec{r}) = \varepsilon_n^{PP}\psi_n^{PP}(\vec{r})$$

solve for $V_{PP}(\vec{r})$

in region where bonding happens $= \psi_n^{AE}(\vec{r})$

ε_n^{AE}

use $V_{PP}(\vec{r})$ for calculations

.UPF files in QE input

https://www.pseudo-dojo.org/

https://pseudopotentials.quantum-espresso.org/legacy_tables/original-qe-pp-library

```
&CONTROL
outdir = './'
pseudo_dir = './'
 /

 &SYSTEM
    ibrav =  1
    A = 12.0
    nat =  2
    ntyp = 1
    ecutwfc = 60.0
 /

 &ELECTRONS
 /

ATOMIC_SPECIES
 H  1.008  H.pz-vbc.UPF

ATOMIC_POSITIONS {angstrom}
 H 0.00 0.00 0.00
 H 0.00 0.00 1.00

K_POINTS {automatic}
  1 1 1 0 0 0
```

LDA

H. pbe

Lecture 3: Practical aspects of DFT

The exchange-correlation functional
Solving the Kohn-Sham equation on a computer
Pseudopotentials

Lab 3: Convergence Tests

https://imperialcollegelondon.github.io/MSE404-MM/

When you log in, you should have a data folder in your home directory, please store your files in this folder.